Types of AI Agents

Understanding Different Kinds of Agents in Artificial Intelligence

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What is an AI Agent?

- An **AI agent** perceives its environment and takes actions to achieve its goals.
- The agent senses its environment through **percepts** (inputs), makes decisions, and performs **actions**.
- The way the agent behaves depends on how it processes its environment and the rules or learning algorithms it follows.

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Types of AI Agents

- Simple Reflex Agent
- Model-Based Reflex Agent
- Goal-Based Agent
- Utility-Based Agent
- Learning Agent

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Simple Reflex Agent

- A **simple reflex agent** reacts to the current situation using predefined rules.
- It doesn't think ahead or remember past actions. It only reacts to the current percept.
- **How it works**: Based on a condition in the environment, it immediately performs a corresponding action.
- **Example**: A robot vacuum that moves forward until it hits an obstacle, then turns and moves in another direction.
- **Rule**: If an obstacle is detected, turn left.

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Model-Based Reflex Agent

- A **model-based reflex agent** remembers past states of the world and uses this memory to make better decisions.
- Unlike the simple reflex agent, it doesn't just react to the current situation; it considers how the world has changed.
- **How it works**: It keeps track of what has happened before and uses this information to update its model of the world.
- **Example**: A thermostat. It doesn't just react to the current temperature but remembers the last temperature and adjusts the heating or cooling accordingly.
- **Rule**: If the temperature is too low (and it was also low earlier), turn on the heater.

Goal-Based Agent

- A **goal-based agent** has specific goals it aims to achieve, and it plans actions to reach those goals.
- Unlike reflex agents, it isn't just reacting. It actively thinks about how to achieve a desired outcome.
- **How it works**: It evaluates its options and chooses the best sequence of actions to achieve its goal.
- **Example**: A GPS navigation system. The system doesn't just react to roads or traffic; it figures out the best route to a destination.
- **Goal**: Find the fastest route from point A to point B.

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Utility-Based Agent

- A **utility-based agent** chooses actions based on a measure of "utility" or satisfaction.
- It doesn't just aim to achieve a goal; it tries to maximize its "happiness" or "satisfaction" from different options.
- **How it works**: It evaluates possible actions according to how much utility they provide and selects the one that gives the highest utility.
- **Example**: A shopping assistant. It helps you choose the best product by evaluating quality, price, reviews, etc., to find the option that maximizes your satisfaction.
- **Utility**: Choose the product that gives the most value (quality + price) based on your preferences.

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Learning Agent

- A **learning agent** improves over time by learning from its experiences.
- It starts with some basic knowledge or rules and then uses feedback from the environment to improve its actions.
- **How it works**: It uses past experiences (correct or incorrect actions) to update its knowledge or strategy, improving its future performance.
- **Example**: A game-playing AI that gets better at playing chess the more it plays, learning new strategies from its wins and losses.
- **Learning Process**: Observe $\rightarrow$ Learn $\rightarrow$ Improve $\rightarrow$ Perform better.

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Summary

- **Simple Reflex Agent**: Reacts to current situations based on predefined rules, with no memory or planning.
- **Model-Based Reflex Agent**: Remembers past states of the world to make better decisions in the present.
- **Goal-Based Agent**: Chooses actions that bring it closer to a specific goal.
- **Utility-Based Agent**: Chooses actions that maximize satisfaction or "utility."
- **Learning Agent**: Learns from past experiences to improve over time.

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Thank You!

Any Questions?

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